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Optimized phase pupil masks for extended depth of field

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Abstract

By properly designing a phase pupil mask to modulate or encode the optical images and then digitally restoring them, one can greatly extend the depth of field and improve image quality. The original works done by Dowski and Cathey introduce the use of a cubic phase pupil mask to extend the depth of field. The theoretical and experimental researches all verified its effectiveness. In this paper, we suggest the use of an exponential phase pupil mask to extend the depth of field. This phase mask has two variable parameters for designing to control the shape of the mask so as to modulate the wave-front more flexible. We employ an optimization procedure based on the Fisher information metric to obtain the optimum values of the parameters for the exponential and the cubic masks, respectively. A series of performance comparisons between these two optimized phase masks in extending the depth of field are then done. The results show that the exponential phase mask provide slight advantage to the cubic one in several aspects.

Keywords: Extended depth of field; Phase pupil mask; Optimization

1. Introduction

An optical imaging system with extended depth of field (DOF) is desired in many applications, for it provides the possibility to acquire sharp images of three-dimensional objects in a single frame as long as they are not thicker than the DOF. An extended DOF can be accomplished in conventional optical imaging system by stopping down the aperture, but this can result in low contrast and noise-dominated images because of decrease in optical flux reaching the image plane. Furthermore, a decrease in aperture size worsens the diffraction limited resolution at the in-focus plane. The conflicts between the DOF and the optical flux as well as the resolution of image do not solved in the conventional optical design concepts, involving the use of absorptive apodization methods [1–6], until the invention of a new imaging modality known as the *wave-front coding* [7–10]. The wave-front coded optical system is a hybrid imaging system that combines a phase modulated optical imaging system and a digital post-processing step. A properly designed phase mask is inserted at the exit pupil plane to encode a three-dimensional scene so as to produce an intermediate image with nearly spatially invariant blur, and a digital image processing is then performed to decode the final image. Since the introduced phase mask only modifies the phase of the received wave-front, the optical power reaching the image space is maximized.

The wave-front coding imaging technique, first proposed by Dowski and Cathey, use a rectangularly separable aperture and a cubic phase pupil mask, of the form $\phi(u) = \alpha u^3$, to extend the DOF. The multiplier α, which describes the maximum phase deviation introduced by the phase mask, is used to balance the magnitude of DOF and the performance of imaging. Small α can not extend the DOF effectively, but too large α can result in a lower modulation transfer function (MTF). It therefore necessary to employ an optimization procedure to obtain a suitable parameter α so as to maximize the system's ability of transfer optical information for a given DOF, or alternatively, maximize the DOF for a given MTF. However, a simple mask with cubic form has only one variable

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parameter to control the wave-front coding behaviour, it is not so flexible in some optical design cases. There have been many works done to search other feasible functional form of phase masks for extending the DOF [11–13]. In this paper, we suggest the "exponential" phase pupil mask, of the functional form $\phi(u) = \beta u \exp(\gamma u^2)$ (where, $\gamma > 0$), to extend the DOF. This form of phase mask has two variable parameters β and γ, together determining the shape of the mask. We term the optical imaging system with an exponential phase mask in the exit pupil plane as the exponential phase modulated (EPM) optical imaging system, known from the cubic phase modulated (CPM) optical imaging system.

2. Optimization of the phase pupil masks for extended depth of field

Consider a phase modulated optical imaging system with a phase mask of the form $\phi(u)$ locating at the exit pupil plane, the system's one-dimensional pupil function $p(u)$ in normalized coordinate, then, is

$$p(u) = \begin{cases} \frac{1}{\sqrt{2}} \exp[\mathrm{j}\phi(u)] & \text{for } |u| \leqslant 1, \\ 0 & \text{for } |u| > 1. \end{cases} \tag{1}$$

The optical transfer function (OTF), as a function of defocus, of this phase modulated optical imaging system can be represented in the following form:

$$\begin{aligned} H(\rho,\psi) &= \int_{-\infty}^{\infty} \{p(u+\rho/2)\exp[\mathrm{j}\psi(u+\rho/2)^2]\} \\ &\quad \times \{p^*(u-\rho/2)\exp[-\mathrm{j}\psi(u-\rho/2)^2]\}\,\mathrm{d}u \\ &= \int_{-\infty}^{\infty} p(u+\rho/2)p^*(u-\rho/2)\exp(\mathrm{j}2\psi\rho u)\,\mathrm{d}u, \end{aligned} \tag{2}$$

where the variable ρ denotes the spatial-frequency and parameter ψ is a measurement of the amount of defocus in units of wavelength.

For an optical imaging system with extended DOF, it requires that the system's OTF is insensitive to the defocus parameters. A convenient mathematical tool to measure the sensitivity of the OTF to focus errors is the Fisher information, which is commonly used to describe the information content of a given signal pertaining to a certain parameter [7]. The Fisher information $J(\psi)$ of the OTF $H(\rho,\psi)$ associated with the defocus parameter ψ is defined by

$$J(\psi) = \int_{-2}^{2} \left| \frac{\partial}{\partial\psi} H(\rho,\psi) \right|^2 \mathrm{d}\rho. \tag{3}$$

The smaller the value of the Fisher information related to the OTF at certain defocus image plane, the lower variation of the OTF to the focus errors near this image plane. For an idea focus-invariant optical imaging system, this Fisher information would be zero for all defocus state. Thus we can employ the integral of the Fisher information

$$I(\psi_0) = \int_{-\psi_0}^{\psi_0} J(\psi)\,\mathrm{d}\psi \tag{4}$$

as a metric to measure the sensitivity of an optical imaging system to focus errors, where $(-\psi_0, \psi_0)$ is the symmetric range of defocus parameters of interest for the physical problem at hand.

Since the shape of the phase pupil mask is the critical optical component to extend the DOF, we naturally wish the parameters to control the shape of this mask are optimal. With the use of Fisher information metric, we can build the following object function:

$$\begin{aligned} &\min_X \quad f(X) = I(X,\psi_0) \\ &\text{subject to} \\ &\qquad X \geqslant X_0; \\ &\qquad g(X) \geqslant \kappa, \end{aligned} \tag{5}$$

where $X \in \mathbb{R}^n$ is the parameters to be optimized. For a cubic phase mask it is the scalar variable α and for the exponential phase mask it becomes the vector variable $[\beta,\gamma]^T$. To avoid trivial solutions, the object function was constrained by two inequalities. The first constraint is the feasible solution space, and the second constraint gives the allowably lowest in-focus MTF values. Therefore, the function $g(X)$ can be defined as

$$g(X) = \int_{-2}^{2} |H(\rho,0)|\,\mathrm{d}\rho. \tag{6}$$

Eq. (5) describes a constrained nonlinear optimization problem. This description of the optimization for optical design in extending the DOF is feasible and reliable, for at one hand by minimizing the total Fisher information of the OTF in the defocus range of interest guarantees the lowest variation of the OTF to defocus, and at the other hand, the constraints added on the objection function assure that the system's MTF would not too low.

By using the Matlab's optimization tool box, we obtain several sets of the optimum value of the parameters for the cubic phase mask given in Table 1 and the exponential phase mask given in Table 2. In the numerical optimization program, we have chose the defocus range as (−30,30) and

Table 1
Cubic phase mask optimum parameters

κ	0.20	0.24	0.28	0.30	0.32	0.36	0.40
α	209.72	141.51	104.19	90.41	79.20	62.26	49.67
$I \times 10^3$	3.72	10.51	23.91	35.07	50.10	95.23	171.15

Table 2
Exponential phase mask optimum parameters

κ	0.20	0.24	0.28	0.30	0.32	0.36	0.40
β	125.70	69.27	54.98	37.41	31.95	33.53	31.42
γ	1.14	1.35	1.28	1.53	1.55	1.25	1.12
$I \times 10^3$	2.05	5.16	11.91	17.29	25.30	50.19	92.78

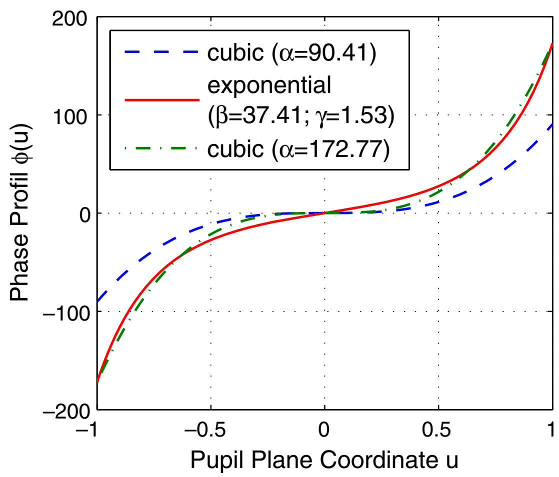

Fig. 1. Phase profiles of three phase masks.

for the exponential phase mask optimization, the lower bound of X is set as $X_0 = [30, 0]^T$. To avoid a local minimum value, a series of initial values were set to find the optimum value. It should be pointed out that the optimum values listed in Tables 1 and 2 are all satisfied the condition $g(X_{\text{optimum}}) = \kappa$. The profiles of the cubic phase mask with parameter $\alpha = 90.41$ and the exponential phase mask with parameters $\beta = 37.41$ and $\gamma = 1.53$ are shown in Fig. 1. The parameters for these two phase masks are all optimized to the equal κ value. The profile of the cubic phase mask with the parameter $\alpha = 172.77$ that has the same maximum phase deviation with the exponential phase mask whose parameters are given above, is also plotted in Fig. 1 for comparison purpose.

3. Performance comparison

3.1. Defocused optical transfer function and defocused point-spread function

3.1.1. Defocused optical transfer function

Before inspecting the defocused OTF of a phase modulated optical imaging system, let us consider first the ambiguity function (AF) of the pupil function of this optical imaging system. The AF $A_p(\rho, x)$ of the pupil function $p(u)$ is defined by

$$A_p(\rho, x) = \int_{-\infty}^{\infty} p(u + \rho/2) p^*(u - \rho/2) \exp(j2\pi x u)\, du. \qquad (7)$$

Comparing Eq. (7) with Eq. (2), it can easily find that the values of the defocused OTF are found from a slice through the origin of the AF of the slope ψ/π. This means that the AF can be used as a polar display of the OTFs with the varying focus errors in a single picture [14]. The gray-scale picture of the AF of an exponential phase pupil function and two cubic phase pupil functions, whose phase profiles are given in Fig. 1, are drawn in Figs. 2–4, respectively. Compared with the AFs of two cubic phase pupil functions, the AF of the exponential phase pupil function has broader range of angles in which the values along the radial lines through the origin of the AF, as a function of angle, are nearly uniform for any possible angles. In addition, the AF of the exponential phase pupil function has

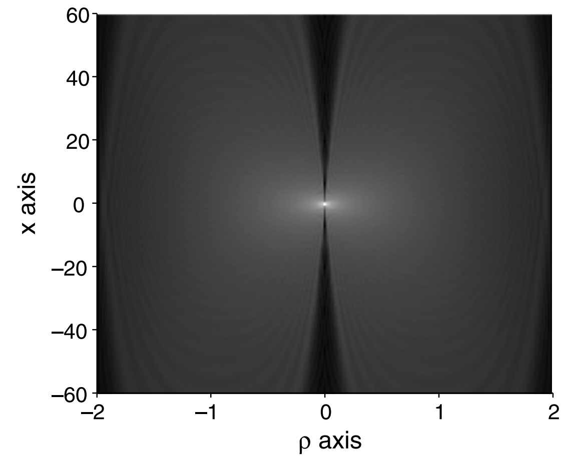

Fig. 2. Gray-scale picture of the AF of the exponential phase pupil function with the parameters $\beta = 37.41$ and $\gamma = 1.53$.

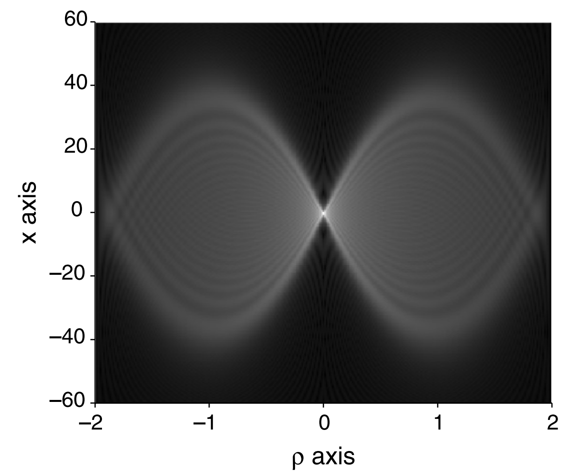

Fig. 3. Gray-scale picture of the AF of the cubic phase pupil function with the parameter $\alpha = 90.41$.

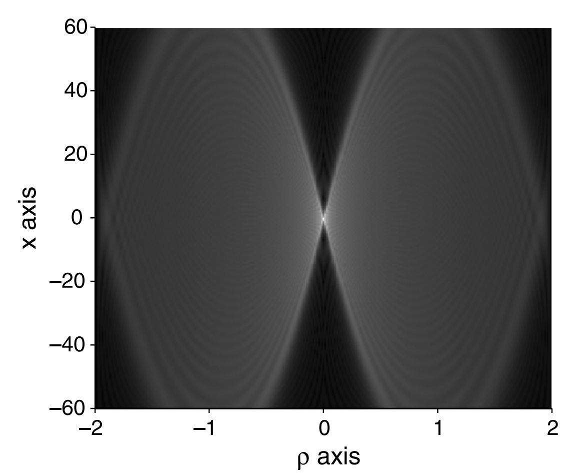

Fig. 4. Gray-scale picture of the AF of the cubic phase pupil function with the parameter $\alpha = 172.77$.

smaller regions where the values approach zero near the maximum spatial-frequency (i.e., near $\rho = \pm 2$). This means that the EPM optical imaging system has a broader bandwidth than the CPM optical imaging system, especially for the extreme defocus state.

The MTF curves of the EPM optical imaging system and two CPM optical systems are plotted in Figs. 5–7. It is apparent that two family of MTF curves of the EPM optical imaging system and the CPM optical imaging systems with parameter $\alpha = 90.41$ have nearly the same κ value, but with less variations of the MTF curves to focus errors for the EPM optical imaging system than the CPM optical imaging system with parameter $\alpha = 90.41$, especially at the higher spatial-frequency part. Increasing the parameter α of the CPM optical imaging system can reduce this variations but result in a lower MTF values as shown in Fig. 7. Additionally, the EPM optical imaging system has a smoother MTF curve, a favorable character in the post digital image processing, than the CPM one.

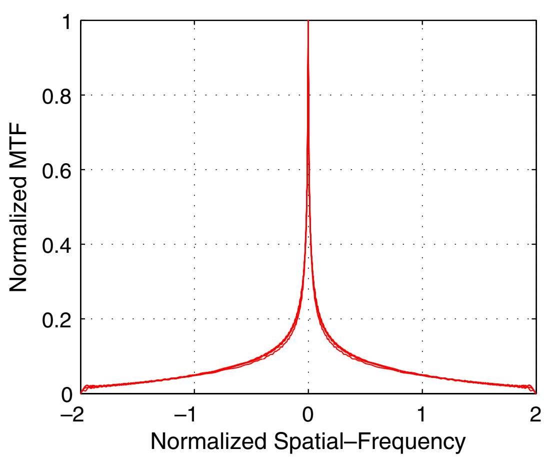

Fig. 5. MTFs of the EPM optical imaging system with the parameters $\beta = 37.41$ and $\gamma = 1.53$, and with defocus ψ of 0, 5, 15, and 30.

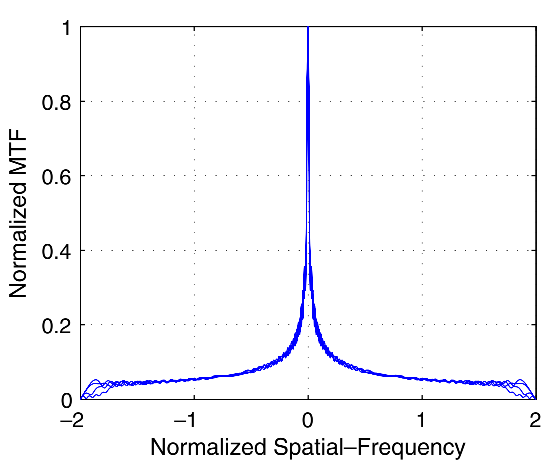

Fig. 6. MTFs of the CPM optical imaging system with the parameter $\alpha = 90.41$, and with defocus ψ of 0, 5, 15, and 30.

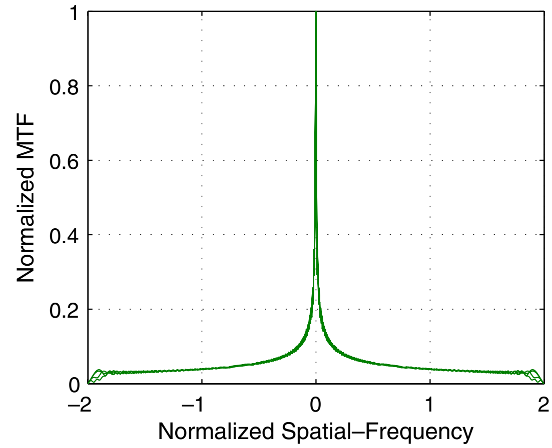

Fig. 7. MTFs of the CPM optical imaging system with the parameter $\alpha = 172.77$, and with defocus ψ of 0, 5, 15, and 30.

3.1.2. Defocused point-spread function

The diffraction-limited PSFs of the EPM optical imaging system are shown in Fig. 8 for four defocus parameters. The radiance of this kind of PSF is mainly concentrated about a peak. On one side of the peak it decays slowly and on the other side it decays rapidly. The PSF of this optical imaging system changes less as the defocus aberrations varying compared with the PSF of the CPM optical imaging system does as shown in Fig. 9. Only at the largest defocus state in the designed DOF, the PSF of the EPM optical imaging system has slight decreasing in the peak value. As is shown in Fig. 10, the increase of the parameter of the CPM optical imaging system can reduce the sensitivity of the PSF to defocus error, but this would reduce the visibility of the images for the decrease of the PSF values.

3.2. Fisher information and Hilbert space angle

In the previous subsection, we have used a qualitative method to study the performance of the exponential phase mask and the cubic one in the extending the DOF. In the following two subsections, we use a numerical criterion to evaluate this two kinds of phase modulated optical imaging systems from the measurement of the Fisher information and the Hilbert space angle.

3.2.1. Fisher information

The Fisher information has employed as an optimization metric above to obtain the optimum parameter values for the phase masks, where the total amount of the Fisher information over a range of DOF is used. The evaluation of the Fisher information for all possible defocus parameters in the range of DOF might help us to see clearer the sensitivity of the optical imaging system to focus errors. The Fisher information as a function of defocus parameter for three phase modulated optical imaging systems are plotted in Fig. 11. It is obvious that the EPM optical imaging system provides a superior performance than the CPM one in extending the DOF for less sensitivity to defocus

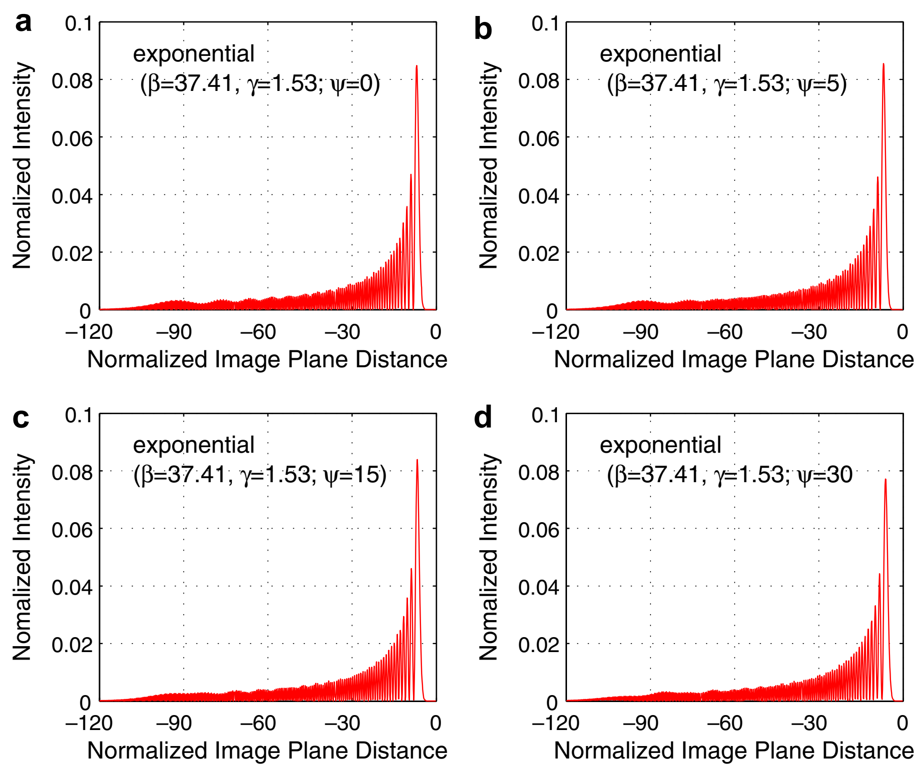

Fig. 8. Defocused PSF of the EPM optical imaging system with the parameters $\beta = 37.41$ and $\gamma = 1.53$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

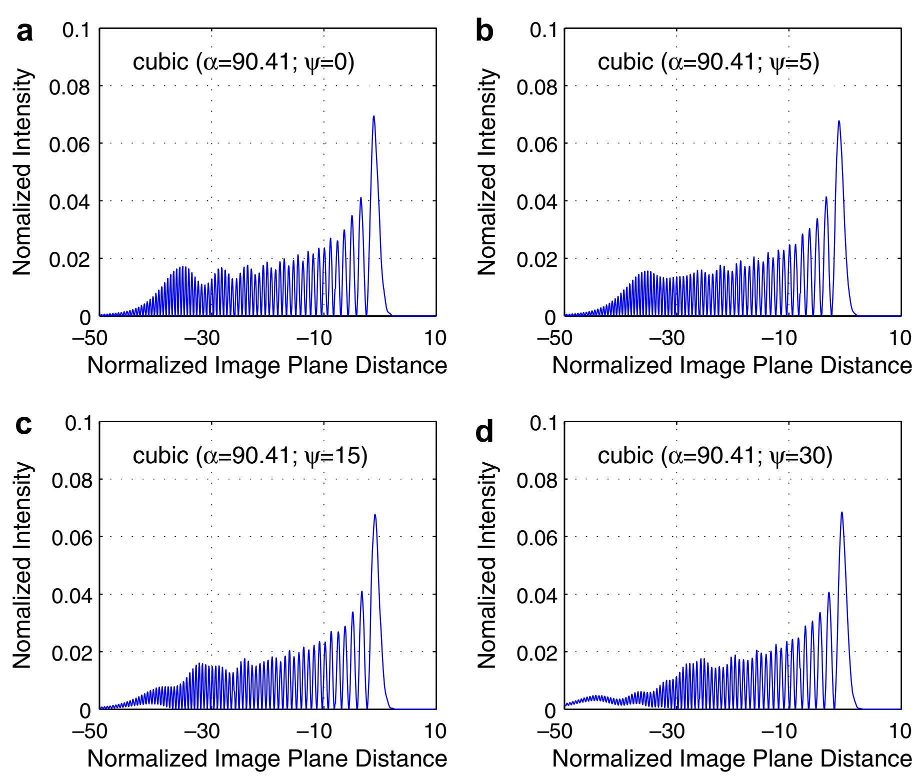

Fig. 9. Defocused PSF of the CPM optical imaging system with the parameter $\alpha = 90.41$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

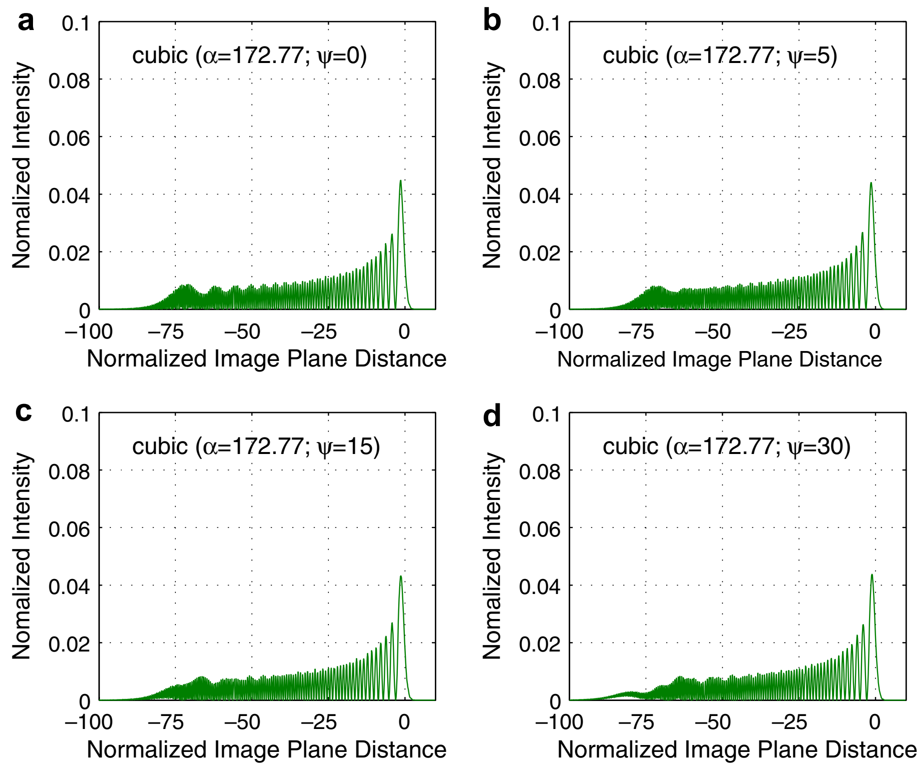

Fig. 10. Defocused PSF of the CPM optical imaging system with the parameter $\alpha = 172.77$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

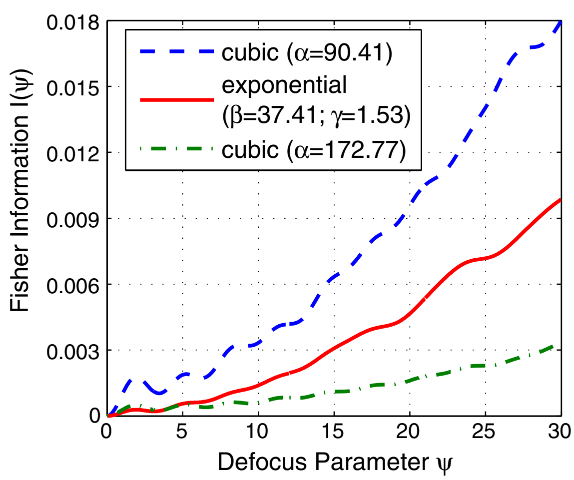

Fig. 11. Fisher information as a function of defocus.

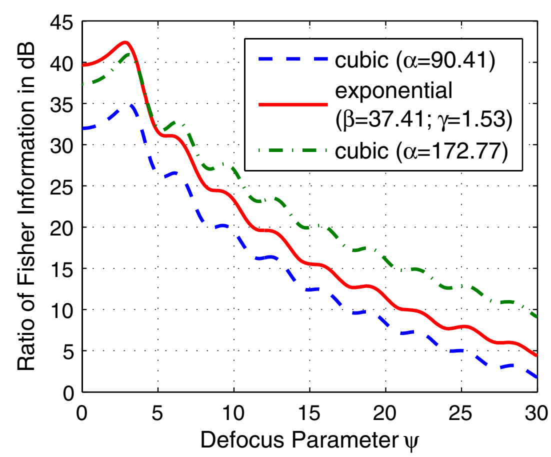

Fig. 12. Ratio of the Fisher information of a conventional optical imaging system to the Fisher information of three phase modulated optical imaging system.

errors, assuming that the parameters of the phase masks are all optimized to make system having the same κ value. From Tables 1 and 2, we can further find that the total Fisher information of the EPM optical system is only about half that of the CPM optical imaging system if the value of κ is the same. To increase the parameter α of the cubic phase mask to be the same as the maximum phase deviation of the exponential phase mask (i.e., $\alpha = \beta e^{\gamma}$) would give an even less sensitivity to defocus errors of the optical imaging system, but this is accomplished by making sacrifice to depress the system's ability to transfer the optical information. The Ratio of the Fisher information of a conventional optical imaging system to the Fisher information of three phase modulated optical imaging system are plotted in Fig. 12. This ratio is a measurement of performance of the phase modulated optical imaging system compared with the conventional optical imaging system. When this ratio is greater than unity, or 0 dB, the theoretical variation with defocus of the conventional optical imaging system exceeds that of the phase modulated optical imaging system. According to the Hopkins criterion, the defocus tolerance of the conventional optical imaging system is $\psi \approx 1$. From Fig. 12, it can be seen that the defocus tolerance of the CPM optical imaging system with parameter $\alpha = 90.41$ is nearly 30 times larger than that of a conventional one, and the EPM optical imaging system with parameters $\beta = 37.41$ and $\gamma = 1.53$ is even larger.

3.2.2. Hilbert space angle

The Hilbert space angle is a convenient mathematical tool to measure the similarity between any two functions [15]. The smaller the angle between the two functions, the more nearly similar are the two functions; the larger the angle between the two functions, the more dissimilar the two functions. The Hilbert space angle between the in-focus PSF (or MTF) and a series of defocus PSF (or MTF) can thus be used as another numerical way to evaluate the performance of an optical imaging systems in extending the DOF. This angle, $0 \leqslant \theta \leqslant \pi/2$, is defined for any defocus parameter ψ as

$$\cos[\theta(\psi)] = \frac{\langle f(x,0) f(x,\psi)\rangle}{\|f(x,0)\|\|f(x,\psi)\|}, \tag{8}$$

where the function f denotes the system's incoherent PSF or MTF. The inner product occurred in Eq. (8) is defined as

$$\langle f(x,0) f(x,\psi)\rangle = \int_{-\infty}^{\infty} f(x,0) f(x,\psi)\,\mathrm{d}x. \tag{9}$$

The norms in the denominator of Eq. (8) are the Hilbert space length for the two functions that are defined as

$$\|f(x,0)\| = \left[\int_{-\infty}^{\infty} f(x,0) f(x,0)\,\mathrm{d}x\right]^{1/2} \tag{10}$$

and

$$\|f(x,\psi)\| = \left[\int_{-\infty}^{\infty} f(x,\psi) f(x,\psi)\,\mathrm{d}x\right]^{1/2}. \tag{11}$$

The Hilbert space angle between the in-focus PSF and defocused PSF with various defocus parameter value ψ is plotted in Fig. 13. As is shown, the EPM optical imaging system shows a more similarity between its in-focus PSF and defocused PSFs compared with the CPM optical imaging system in a large defocus range if the parameters are all optimized to the same κ value. The exception appear in the large defocus state (about $\psi \geqslant 27$) is due to the slight decrease of the peak value of the defocused PSF of the EPM optical imaging system. For the CPM optical imaging system, the only way to make the defocused PSF more similar to the in-focus one is to increase the phase mask parameter α but this is trivial as has been explained above. Fig. 14 shows the Hilbert space angle between the in-focus MTF and defocused MTF with varying defocus parameter value ψ. It is not unexpected that the EPM optical imaging system presents higher similarity between its in-focus MTF and defocused MTF for all defocus parameters than that the two CPM optical imaging systems do, because the MTF of the EPM optical imaging system exhibits less variation along the spatial-frequency coordinate than that of

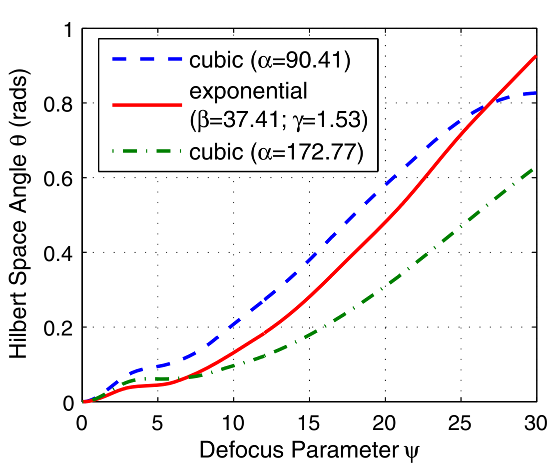

Fig. 13. Hilbert space angles between the defocused PSF and the in-focus PSF.

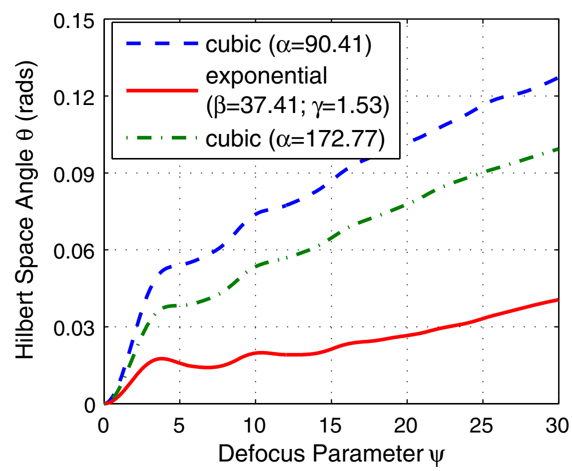

Fig. 14. Hilbert space angles between the defocused MTF and the in-focus MTF.

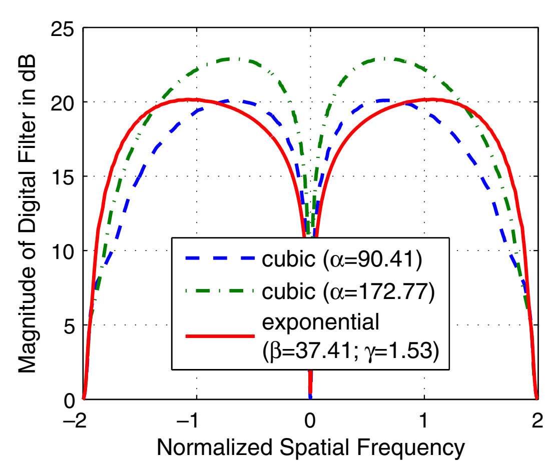

Fig. 15. Magnitude of the digital filter transfer function for three phase modulated optical imaging systems.

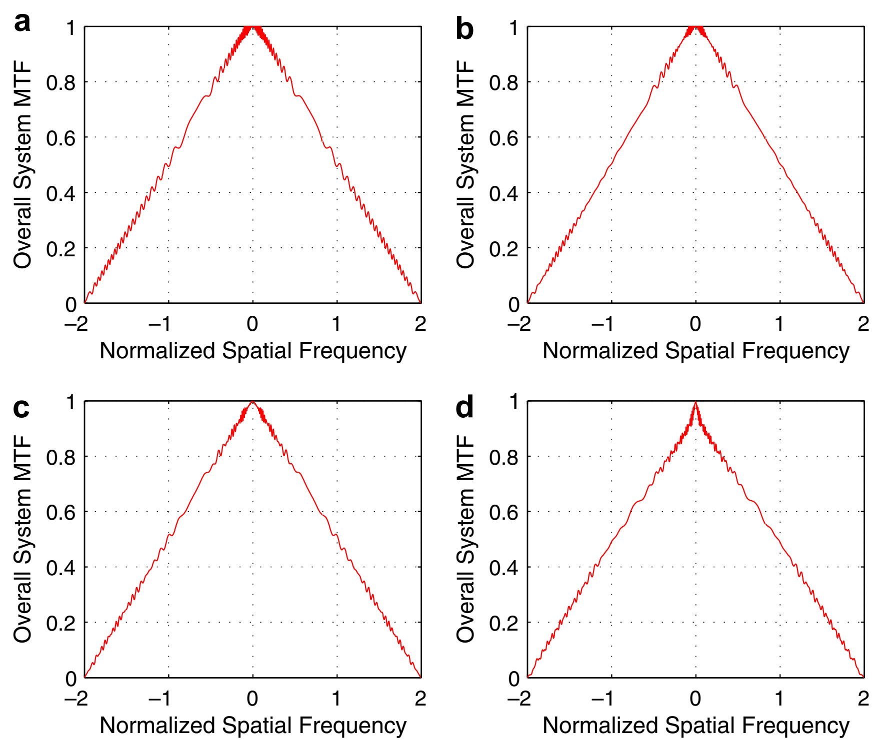

Fig. 16. The overall system's MTFs of the EPM optical imaging system with the parameters $\beta = 37.41$ and $\gamma = 1.53$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

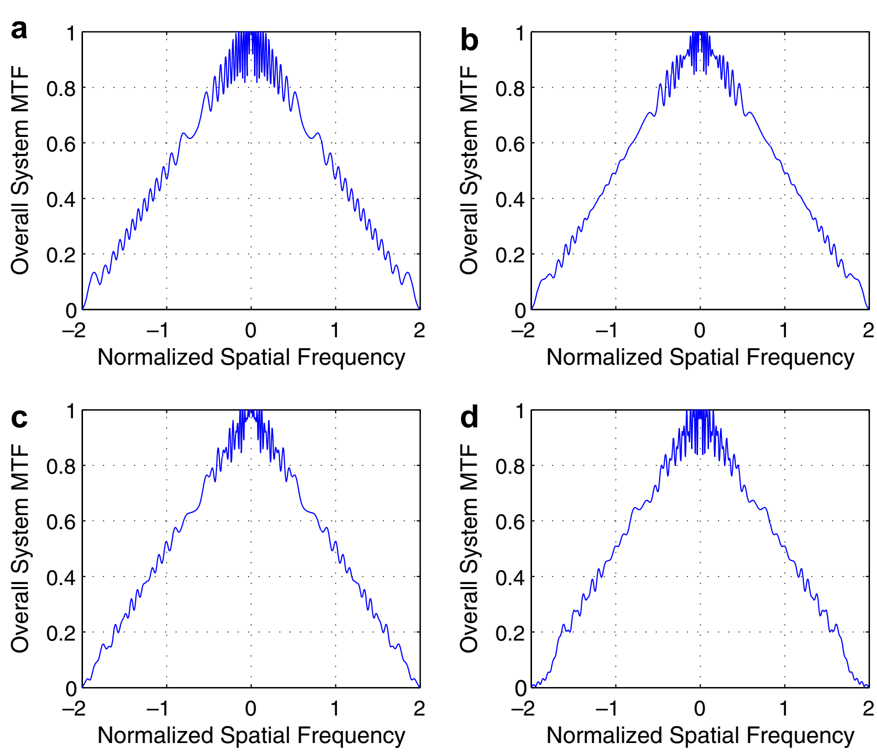

Fig. 17. The overall system's MTFs of the CPM optical imaging system with the parameter $\alpha = 90.41$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

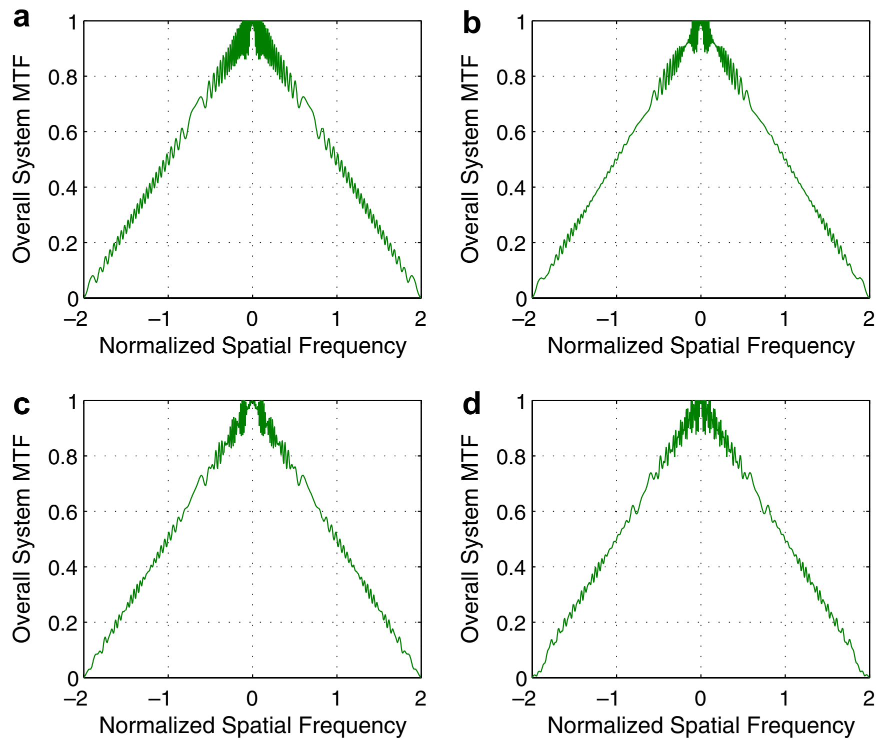

Fig. 18. The overall system's MTFs of the CPM optical imaging system with the parameter $\alpha = 172.77$, and with defocus parameter: (a) $\psi = 0$, (b) $\psi = 5$, (c) $\psi = 15$ and (d) $\psi = 30$.

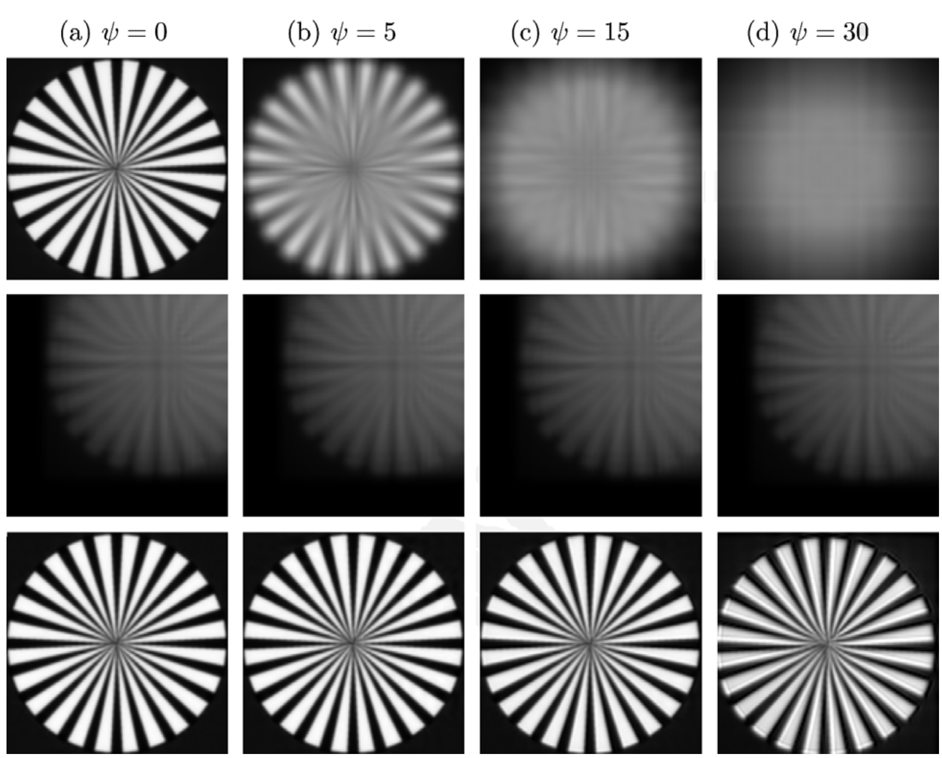

Fig. 19. Simulated images of a spoke pattern. The first row is the clear rectangular aperture; The second and third rows are the intermediate and final images of the EPM optical imaging system with parameters $\beta = 37.41$ and $\gamma = 1.53$.

the CPM one. This property is one of the advantages that the EPM optical imaging system owns.

3.3. Image restoration and simulated images

All the CPM and EPM optical imaging systems change the phase form of the incident wave-front, making the image acquired by these optical imaging systems unrecognizable. A digital image processing is then required to restore the final image. In addition, the loss of visibility of the intermediate images can also be enhanced by a digital image processing. Since these two optical imaging system deliver nearly the same blurred images with varying defocus, a single digital filter can be used to restore the final image. Fig. 15 shows the simple inverse digital filters, which are designed in a least mean square method to approximate an idea triangular OTF over a range of defocus, for the EPM and two CPM optical imaging systems. The higher maximum value of the magnitude of the digital filter for the CPM optical imaging system with parameter $\alpha = 172.77$ means that it requires more signal-to-noise-ratio or dynamic-range premium for the image detector compared with the other two optical imaging systems. The overall system's MTFs after filtering are plotted in Figs. 16–18 for three phase modulated optical imaging system. Compared with the two CPM optical imaging systems, the EPM optical imaging system's overall MTF more resembles the idea triangular OTF for all the defocus parameters, owing to the smoothness of the EPM optical imaging system's MTF curves.

To demonstrate the performance of the EPM optical imaging system in extending the DOF, we compare four sets of simulated images of a spoke target for different defocus parameters as shown in Figs. 19 and 20. Compared with the conventional optical imaging system with rectangular apertures, the EPM and two CPM optical imaging systems all provide highly invariant intermediate images and approximately diffraction-limited final images for any focus errors. The shift of images in the image plane for the EPM optical imaging system, that caused by large shift of PSF, can be complimented by adding a linear phase term on the exponential one. The tiny shades appeared in the final spoke images for the CPM optical imaging system with parameters $\alpha = 90.41$ might due to the digital filter

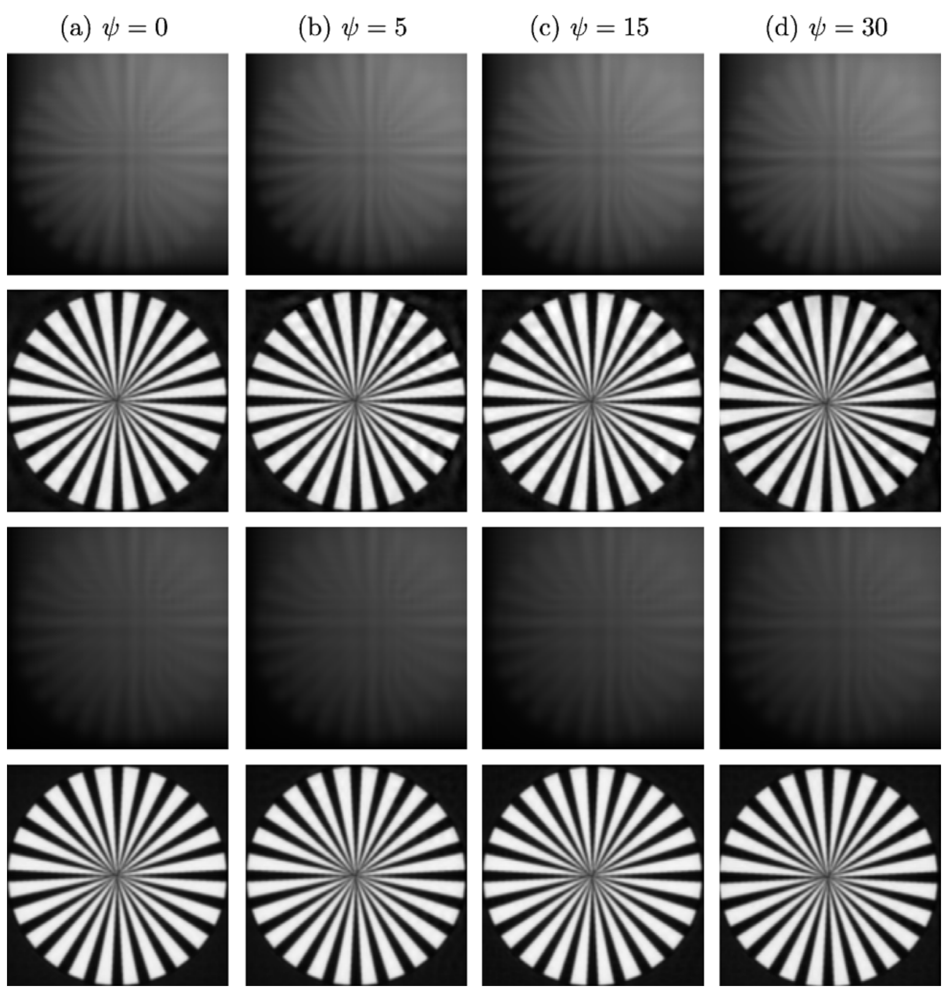

Fig. 20. Simulated images of a spoke pattern. The first two rows are the intermediate and final images of the CPM optical imaging system with parameter $\alpha = 90.41$. The final two rows are the intermediate and final images of the CPM optical imaging system with parameter $\alpha = 172.77$.

can not totally cancel the phase term. The large parameter α (e.g. $\alpha = 172.77$) CPM optical image system can reduce this phenomenon, but result in a low visibility of the intermediate images and thus need more signal-to-noise-ratio or dynamic-range premium of the image detector.

4. Conclusion

We have proposed to use an exponential phase pupil mask to modify the phase of incident wave-front to produce an incoherent optical imaging system with extended DOF. We use an optimization program to obtain the optimum value of the parameters for this exponential phase mask and the cubic one. The performance in extending the DOF is then compared between this two phase masks by a series of evaluations. It is shown that the exponential phase mask provide a slight advantage to the cubic one in several aspects.

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